

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Module - II

PRACTICAL - 6

(Backend Development Using Node.js, Express, and MongoDB)

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Class: TYIT	Batch: 1
Date of Assignment: 09-09-2026	Date/Time of Submission: 09-09-2026

AIM :- Node.js and Express Fundamentals

A) Install Express and create a simple server.

STEP 1 :- Create a folder and open it in VS Code and then inside the terminal write `npm init -y` that will create the `package.json` and then type the following command in order to install the Express

```
PS C:\Users\mvluc\Desktop\Expresspractical> npm install express

added 68 packages, and audited 69 packages in 5s

27 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

STEP 2 :- Create a `server.js` file and write this code inside it

CODE :-

```
const express = require("express");
const app = express();
const PORT = 3000;
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

OUTPUT :-

```
Problems Output Debug Console Terminal Ports
C:\Program Files\nodejs\node.exe .\server.js
Server is running on http://localhost:3000
```

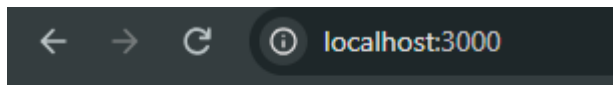
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B) Display a message on the browser using the server response.

CODE :-

```
const express = require("express");
const app = express();
const PORT = 3000;
app.get("/", (req, res) => {
  res.send("Welcome to Express Server");
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

OUTPUT :-



Welcome to Express Server

C) Create different routes in Express.

CODE :-

```
const express = require("express");
const app = express();
const PORT = 3000;
// Home Route
app.get("/", (req, res) => {
  res.send(`
<h1>MVLU College Management Server</h1>
<p>Welcome to our college website.</p>
`);
});
// Students Route
app.get("/students", (req, res) => {
  res.send(`
<h1>Student List</h1>
<ul>
<li>Sagar Kandari</li>
<li>Som mishra</li>
<li>Ashwin</li>
<li>Vijay</li>
</ul>
`);
});
// Faculty Route
app.get("/faculty", (req, res) => {
  res.send(`
<h1>Faculty List</h1>
`);
});
```

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```
<ul>
<li>Sneha Gokarnkar - Information Technology</li>
<li>Rohini Bhosale - Information Technology</li>
<li>Sumit Tripathi - Information Technology</li>
<li>Chetana Tanavade - Information Technology</li>
</ul>
`);
});
// Courses Route
app.get("/courses", (req, res) => {
res.send(`
<h1>Available Courses</h1>
<ul>
<li>B.Sc. Information Technology</li>
<li>B.Sc. Computer Science</li>
<li>B.Sc. Biotechnology</li>
<li>B.A. Multimedia and Mass Communication</li>
</ul>
`);
});
// Contact Route
app.get("/contact", (req, res) => {
res.send(`
<h1>Contact Us</h1>
<p>Email: info@mvlucollege.in</p>
<p>Phone: 9876543210</p>
<p>Address: Andheri, Mumbai</p>
`);
});
// Start Server
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);
});
```

OUTPUT :-

MVLU College Management Server

Welcome to our college website.

Student List

- Sagar Kandari
- Som Mishra
- Ashwin
- Vijay

Available Courses

- B.Sc. Information Technology
- B.Sc. Computer Science
- B.Sc. Biotechnology
- B.A. Multimedia and Mass Communication

Faculty List

- Sneha Gokarnkar - Information Technology
- Rohini Bhosale - Information Technology
- Sumit Tripathi - Information Technology
- Chetana Tanavade - Information Technology

D) Send a JSON response from the server.

CODE :-

```
const express = require("express");
const app = express();
const PORT = 3000;
app.get("/", (req, res) => {
  res.send("Welcome to MVLU College Management Server");
});
app.get("/students", (req, res) => {
  const students = [
    {
      id: 1,
      name: "Sagar Kandari",
      course: "B.Sc. IT",
      marks: 85
    },
    {
      id: 2,
      name: "som mishra",
```

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```
course: "B.Sc. IT",  
marks: 92  
},  
{  
id: 3,  
name: "Ashwin",  
course: "B.Sc. CS",  
marks: 78  
}  
];  
res.json(students);  
});  
app.listen(PORT, () => {  
console.log(`Server is running on http://localhost:${PORT}`);  
});
```

OUTPUT :-

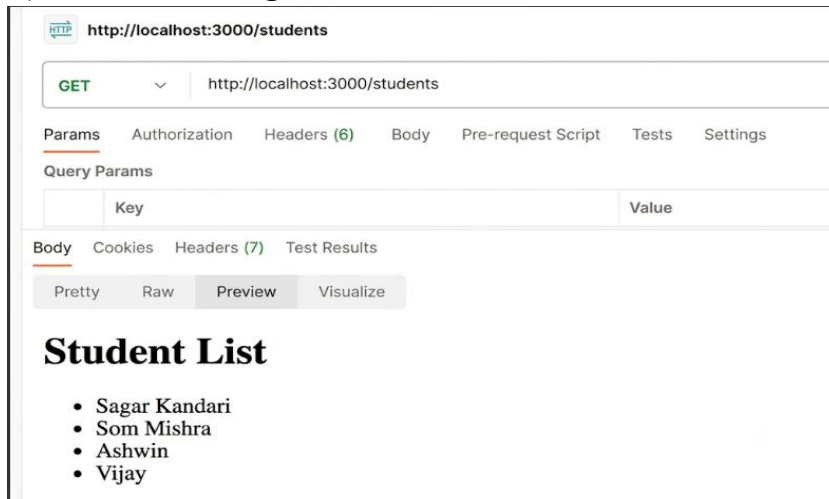


The screenshot shows a web browser window with a dark theme. The address bar at the top shows 'localhost:3000'. The main content area displays a JSON array of three student objects. The first object has id 1, name 'Sagar Kandari', course 'B.Sc. IT', and marks 85. The second object has id 2, name 'Som Mishra', course 'B.Sc. IT', and marks 92. The third object has id 3, name 'Ashwin', course 'B.Sc. CS', and marks 78. The JSON is formatted with syntax highlighting. Above the JSON, there is a 'Pretty-print' button with a checkmark icon. Below the JSON, there is a navigation bar with back, forward, and refresh icons, and a status icon.

```
[  
  {  
    "id": 1,  
    "name": "Sagar Kandari",  
    "course": "B.Sc. IT",  
    "marks": 85  
  },  
  {  
    "id": 2,  
    "name": "Som Mishra",  
    "course": "B.Sc. IT",  
    "marks": 92  
  },  
  {  
    "id": 3,  
    "name": "Ashwin",  
    "course": "B.Sc. CS",  
    "marks": 78  
  }  
]
```

Welcome to MVLU College Management Server

E) Test routes using browser or Postman.



HTTP **GET** http://localhost:3000/students

Params Authorization Headers (6) Body Pre-request Script Tests Settings

Query Params

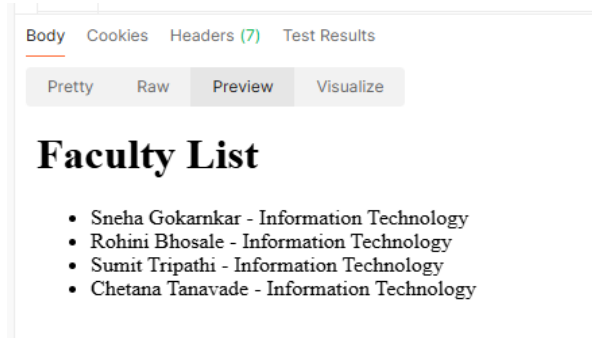
Key	Value
-----	-------

Body Cookies Headers (7) Test Results

Pretty Raw Preview Visualize

Student List

- Sagar Kandari
- Som Mishra
- Ashwin
- Vijay

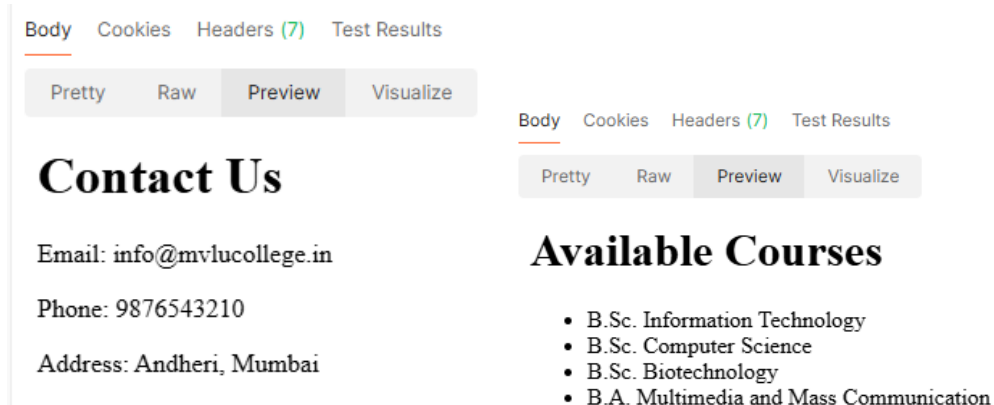


Body Cookies Headers (7) Test Results

Pretty Raw Preview Visualize

Faculty List

- Sneha Gokarnkar - Information Technology
- Rohini Bhosale - Information Technology
- Sumit Tripathi - Information Technology
- Chetana Tanavade - Information Technology



Body Cookies Headers (7) Test Results

Pretty Raw Preview Visualize

Contact Us

Email: info@mvlucollege.in

Phone: 9876543210

Address: Andheri, Mumbai

Available Courses

- B.Sc. Information Technology
- B.Sc. Computer Science
- B.Sc. Biotechnology
- B.A. Multimedia and Mass Communication

F) Run the server and observe the output in the console.

```
PS C:\Users\mvluc\Desktop\Expresspractical> node server.js
Server is running on http://localhost:3000
```